

Problem

A $2\text{-}\Omega$ resistor has $i(t) = 2e^{-t}u(t)$ A. What percentage of the total energy is in the frequency band $-4 < \omega < 4$ rad/s?

Step-by-step solution

Step 1 of 5

Given that $i(t) = 2e^{-t}u(t)$ A, then

$$I(\omega) = \frac{2}{1 + j\omega}$$

$$|I(\omega)|^2 = \left| \frac{2}{1 + j\omega} \right|^2$$

$$|I(\omega)|^2 = \frac{4}{1 + \omega^2}$$

Step 2 of 5

The total energy dissipated by the $2\text{-}\Omega$ resistor is

$$W_{2\Omega} = 2 \left[\frac{1}{\pi} \int_0^{\infty} |I(\omega)|^2 d\omega \right]$$

$$W_{2\Omega} = \frac{2}{\pi} \int_0^{\infty} \frac{4}{1 + \omega^2} d\omega$$

$$W_{2\Omega} = \frac{8}{\pi} \int_0^{\infty} \left(\frac{1}{1 + \omega^2} \right) d\omega$$

$$W_{2\Omega} = \left(\frac{8}{\pi} \right) \left[\tan^{-1} \omega \right]_0^{\infty}$$

Step 3 of 5

$$W_{2\Omega} = \left(\frac{8}{\pi} \right) \left[\tan^{-1} \infty - \tan^{-1} 0 \right]$$

$$W_{2\Omega} = \left(\frac{8}{\pi} \right) \left[\frac{\pi}{2} - 0 \right]$$

$$W_{2\Omega} = 4 \text{ J}$$

The energy in the frequency band $-4 < \omega < 4$ rad/sec is

$$W = \frac{2}{\pi} \int_0^4 |I(\omega)|^2 d\omega$$

Step 4 of 5

$$W = \frac{2}{\pi} \left[\int_0^4 \frac{4}{1+\omega^2} d\omega \right]$$

$$W = \frac{8}{\pi} \left[\int_0^4 \left(\frac{1}{1+\omega^2} \right) d\omega \right]$$

$$W = \left(\frac{8}{\pi} \right) \left[\tan^{-1} \omega \right]_0^4$$

$$W = \left(\frac{8}{\pi} \right) \left[\tan^{-1} 4 - \tan^{-1} 0 \right]$$

Step 5 of 5

$$W = \left(\frac{8}{\pi} \right) \left[\frac{75.964^\circ}{180^\circ} \pi - 0 \right]$$

$$W = 3.3762 \text{ J}$$

Its percentage of the total energy is

$$\frac{W}{W_{2\Omega}} \times 100 = \frac{3.3762}{4} \times 100$$

$$\therefore \frac{W}{W_{2\Omega}} \times 100 = 84.4 \%$$